Scratch Mini Project – Cricket Game Explanation

1. Introduction

This Scratch mini project is a simple Cricket Batting Game. In this game, the player controls a batter. A ball falls from the top of the screen, and when it touches the bat, the player scores runs. If the ball is hit properly, the word 'SIX' appears on the screen. The game also keeps track of score and hit count.

2. Sprites Used in the Project

• Batter Sprite – Controls the batting animation.

• Ball Sprite – Falls from the top and detects collision.

• SIX Sprite – Displays the text 'SIX' when the ball is hit.

3. Variables Used

• Score – Stores the total runs scored by the player.

• Hit – Counts how many times the bat hits the ball.

4. Batter Script Explanation

When the space key is pressed:
• Score is set to 0.
• Hit is set to 0.
• The batter changes costumes (batter-a, batter-b, batter-c, batter-d) quickly.
• This creates a batting animation effect.

5. Ball Script Explanation

When the green flag is clicked:
• The ball goes to the top of the screen at a random X position.
• It continuously falls downward using change y by -9.
• If the ball touches the batter:

- It broadcasts 'SIX'.
- Plays a pop sound.
- Changes direction and moves fast to simulate a hit.
- Score can be increased.

• When the ball reaches the bottom, it restarts from the top.

6. SIX Sprite Script Explanation

When it receives the broadcast 'SIX':

• The SIX sprite shows on the screen.
• It waits for 5 seconds.
• Then it hides again.

This gives a visual effect when the player hits a six.

7. Game Logic Summary

• The ball continuously falls from the top.
• The batter swings when the space key is pressed.
• If collision happens, score increases and SIX appears.
• The game uses loops, conditions, variables, broadcasts, and animation.

8. Scratch Concepts Used

• Events (When green flag clicked, When key pressed)

• Control blocks (Forever, Repeat, If condition)

• Motion blocks (Change y, Move steps, Go to position)

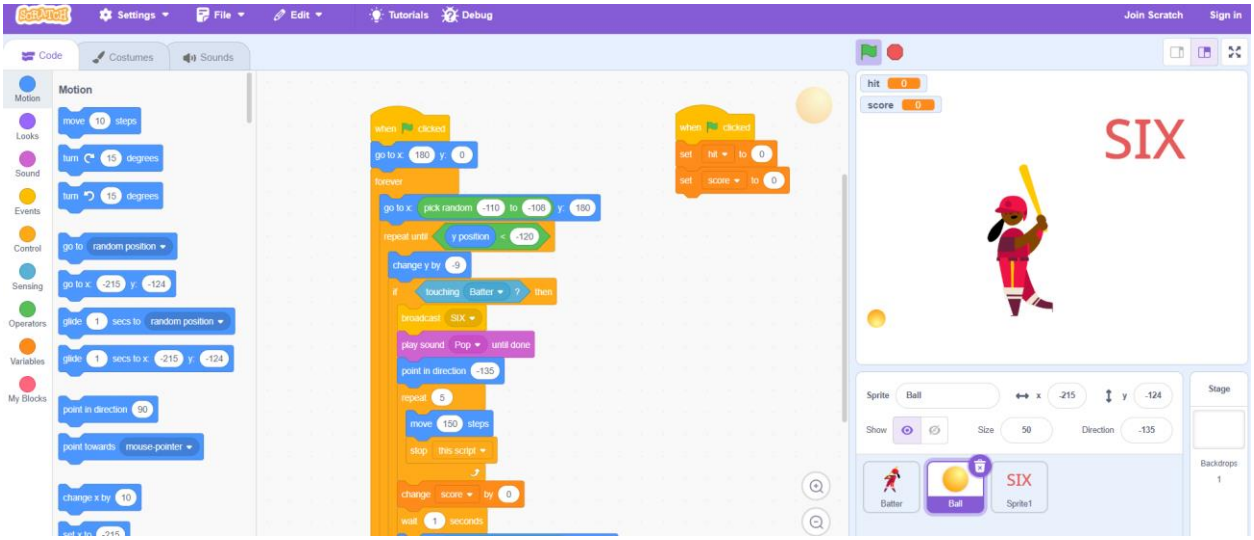
• Variables (Score, Hit)

• Broadcast messages

• Costume switching for animation

9. Conclusion

This Scratch Cricket Game demonstrates the use of animation, collision detection, variables, and broadcasting. It is a beginner-friendly project that shows how games can be created using basic programming logic in Scratch.


Settings
File
Edit
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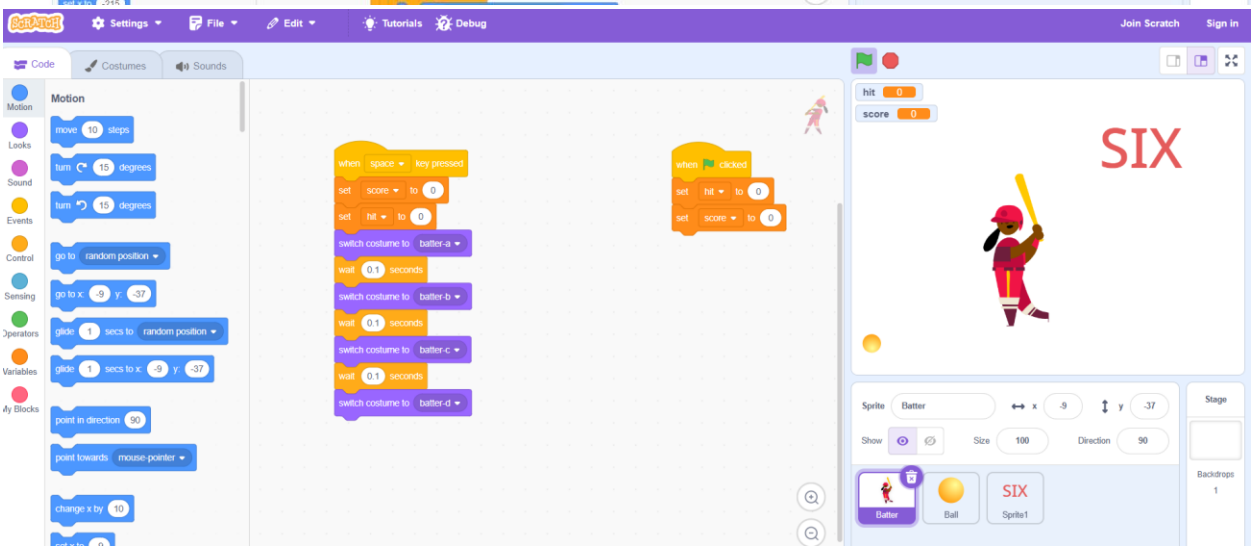
Sign in
Code
Costumes
Sounds
Motion
Looks
Sound
Events
Control
Sensing
Operators
Variables
My Blocks
move 10 steps
turn 15 degrees
turn 15 degrees
go to random position
go to x: -215 y: -124
glide 1 secs to random position
glide 1 secs to x: -215 y: -124
point in direction 90
point towards mouse-pointer
change x by 10
when clicked
go to x: 180 y: 0
forever
go to x: pick random -110 to -108 y: 180
repeat until y position < -120
change y by -9
if touching Batter ? then
broadcast SIX
play sound Pop until done
point in direction -135
repeat 5
move 150 steps
stop this script
change score by 0
wait 1 seconds
when clicked
set hit to 0
set score to 0
hit 0
score 0
SIX
Sprite Ball
x -215
y -124
Show
Size 50
Direction -135
Stage
Backdrops 1
Batter
Ball
SIX
Sprite1


Settings
File
Edit
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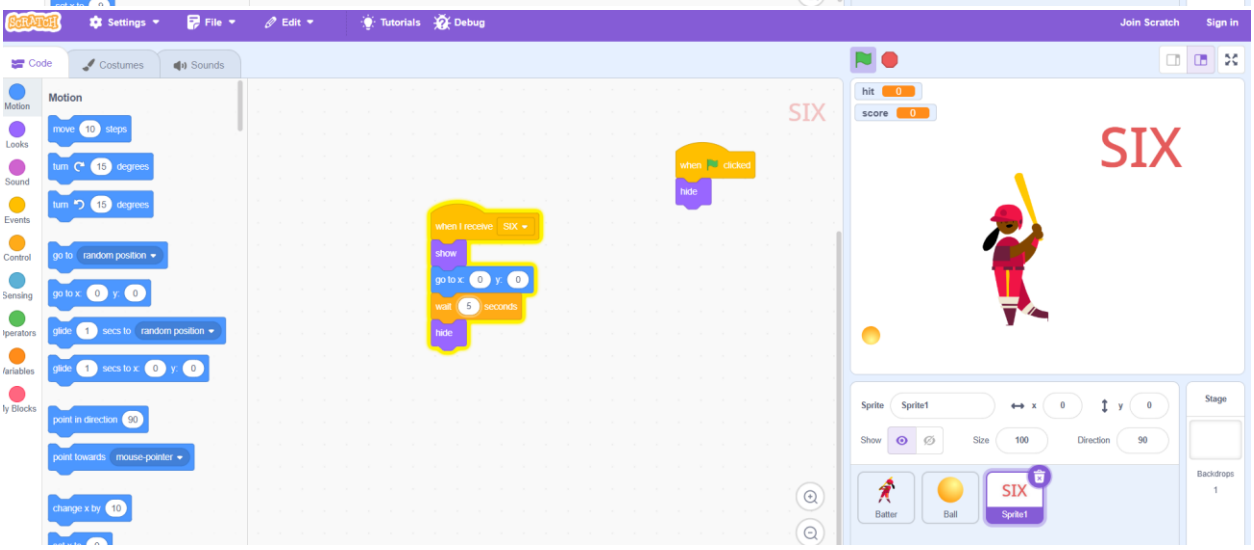
Debug
Join Scratch
Sign in
Code
Costumes
Sounds
Motion
move 10 steps
turn 15 degrees
turn 15 degrees
go to random position
go to x: -9 y: -37
glide 1 secs to random position
glide 1 secs to x: -9 y: -37
point in direction 90
point towards mouse-pointer
change x by 10
when space key pressed
set score to 0
set hit to 0
switch costume to batter-a
wait 0.1 seconds
switch costume to batter-b
wait 0.1 seconds
switch costume to batter-c
wait 0.1 seconds
switch costume to batter-d
when clicked
set hit to 0
set score to 0
hit 0
score 0
SIX
Sprite Batter
x -9
y -37
Show
Size 100
Direction 90
Stage
Backdrops 1
Batter
Ball
SIX
Sprite1


Settings
File
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Debug
Join Scratch
Sign in
Code
Costumes
Sounds
Motion
move 10 steps
turn 15 degrees
turn 15 degrees
go to random position
go to x: 0 y: 0
glide 1 secs to random position
glide 1 secs to x: 0 y: 0
point in direction 90
point towards mouse-pointer
change x by 10
SIX
when I receive SIX
show
go to x: 0 y: 0
wait 5 seconds
hide
when clicked
hide
hit 0
score 0
SIX
Sprite Sprite1
x 0
y 0
Show
Size 100
Direction 90
Stage
Backdrops 1
Batter
Ball
SIX
Sprite1